

Article

Digital Twin Framework for Structural Health Monitoring of Transmission Towers: Integrating BIM, IoT and FEM for Wind–Flood Multi-Hazard Simulation

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Abstract

Transmission towers, as critical infrastructure in power systems, are frequently threatened by multiple hazards such as strong winds and flood scour. Traditional structural health monitoring methods face limitations in data feedback timeliness and mechanical interpretation, making real-time condition awareness and early warning under disaster scenarios challenging. To address these issues, this paper proposes a digital twin framework for transmission tower structures, integrating Building Information Modeling (BIM), Internet of Things (IoT) technology, and the Finite Element Method (FEM) for structural health monitoring and visual warning under wind loads and flood scour effects. The framework achieves cross-platform collaboration through the FEM Open Application Programming Interface (OAPI) and Python scripts. In the physical domain, fluctuating wind loads are simulated based on the Davenport spectrum, flood scour depth is modeled using the HEC-18 formulation, and foundation constraint degradation is represented through nonlinear spring stiffness reduction. In the FEM domain, dynamic time-history analyses are conducted to obtain structural responses. In the BIM domain, a three-level warning mechanism based on stress change rate (ΔR) is established to achieve intuitive rendering and dynamic feedback of structural damage. A 44.4 m high latticed angle steel tower is employed as the case study for validation. Results demonstrate that the simulated wind spectrum closely matches the theoretical target spectrum, confirming the validity of the load input. A critical scour evolution threshold of 40% is identified, beyond which the first two natural frequencies exhibit nonlinear decay with a maximum reduction of 80.9%. Non-uniform scour induces significant load transfer, with axial forces at leeside nodes increasing from 27 kN to 54 kN. During the 0–60 s wind loading process, BIM visualization accurately captures the full stress evolution from the tower base to the upper structure, showing excellent agreement with FEM results. The proposed framework establishes a closed-loop interaction mechanism of “physical sensing–digital simulation–visual warning”, effectively enhancing the timeliness and interpretability of structural health monitoring for transmission towers under multiple hazards, providing an innovative approach for intelligent disaster prevention in power infrastructure.



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Keywords: digital twin; transmission tower; structural health monitoring; Building Information Modeling (BIM); Internet of Things (IoT); Finite Element Method (FEM)

1. Introduction

Transmission towers serve as lifelines ensuring the normal operation of national power systems, and their safety and stability directly affect urban functioning. Angle steel towers, as a common type of transmission tower, are constructed from angle steel sections, with main members, bracing members, and auxiliary members forming a spatial truss structure connected by bolts. Due to their ability to fully utilize material compressive performance and their relatively small slenderness ratios, such latticed structures are widely used in urban and mountainous transmission lines, particularly suitable for voltage levels of 110–500 kV. Angle steel towers often exceed 30 m in height, classifying them as typical high-rise structures [1]. In recent years, wind and flood disasters have frequently occurred in urban and mountainous areas. Wind load, as the primary controlling load, often leads to the collapse or severe deformation of angle steel towers [2–7]; meanwhile, tower foundation failures caused by flood scour are also common, as documented in references [8–10]. Once a transmission tower is damaged, power grid system failures can easily trigger blackouts, not only seriously threatening people's lives and property but also hindering national industrial development. Thus, transmission towers face the dual threat of strong winds and floods.

With the increasing frequency of disasters, higher requirements have been placed on the wind and flood resistance of transmission tower structures, making structural health monitoring an increasingly effective approach. Due to the high cost, destructive nature, and irreversibility of physical experiments, structures can typically only be equipped with physical sensors to measure mechanical response parameters such as strain and displacement. However, in actual post-disaster scenarios, only post-event analysis is usually possible, while pre-disaster prevention and real-time monitoring have long been difficult to achieve. To optimize monitoring schemes before actual deployment, Internet of Things (IoT) technology has gained widespread attention and application in scientific research [11–15]. By deploying virtual sensors in structures, the functions of physical sensors can be simulated, outputting mechanical parameters at terminals. Compared to physical sensors, virtual sensors have extremely low costs, enabling large-scale virtual deployment and data feedback, offering significant advantages for pre-disaster monitoring. Additionally, Building Information Modeling (BIM) technology is also regarded as a reliable tool for structural health monitoring.

In the interaction process between the IoT and BIM, traditional workflows typically use BIM as a visualization interface and sensor feedback platform, while the IoT converts physical parameters into output parameters, packages them for upload to cloud servers, and then relies on manual judgment to determine whether structural damage has occurred. Although this approach has improved work efficiency to some extent, several non-negligible issues remain. First, traditional IoT systems typically serve only as passive data acquisition and transmission terminals, lacking the ability to autonomously interpret the mechanical mechanisms behind abnormal data [16–19]. This disconnects the monitoring data from the physical relationships with specific disaster scenarios (such as flood scour and strong winds), making fault diagnosis heavily dependent on manual post-event analysis and unable to achieve real-time traceability of disaster causes. Second, current monitoring systems are isolated from traditional structural mechanics analysis programs, making it difficult to achieve dynamic interaction with professional structural analysis software such as SAP2000, Abaqus, and Ansys, thus preventing real-time simulation.

Therefore, existing structural health monitoring technologies urgently need upgrading. By achieving dynamic interaction between FEM and IoT/BIM, the limitations of traditional monitoring approaches—namely, the disconnection between data and mechanical analysis—can be effectively overcome. The deep integration of IoT monitoring technology

with structural simulation, by constructing a digital twin system with dynamic perception and real-time feedback capabilities, demonstrates broad application prospects. Applying this technology to transmission tower structures in national power systems not only enables real-time health monitoring under disaster scenarios but also allows for the separate simulation of mechanical effects of wind loads and flood scour in finite element programs, coordinated with IoT technology for terminal visual feedback through BIM models. This novel power grid monitoring framework provides a feasible pathway for achieving genuine “digital twins” in power infrastructure, effectively promoting the intelligent and high-quality development of power grids.

To fulfill the above objectives, this paper proposes a digital twin framework for transmission tower systems, aiming to enable real-time monitoring and visual warning of structural health under wind and flood disasters. The framework seamlessly integrates Building Information Modeling (BIM), Internet of Things (IoT) technology, and the Finite Element Method (FEM) through the Open Application Programming Interface (OAPI) and Python (version 3.13) scripts. Operating as a near real-time, simulation-driven system, the framework accounts for computational latency; in this workflow, virtual sensor outputs automatically trigger FEM re-analysis. The total latency is approximately 3–5 min per cycle, with synchronization maintained through a timestamp-matching script to ensure data consistency between hazard inputs and structural responses. In the physical domain, fluctuating wind loads described by the Davenport spectrum and flood scour effects simulated using the HEC-18 model are represented. In the FEM domain, dynamic time-history analyses are conducted to obtain structural responses. In the BIM domain, a three-level warning mechanism based on stress change rates is established to provide intuitive rendering and dynamic feedback of structural damage. Through validation on a 44.4-m-high angle steel tower, the framework effectively simulates structural behavior under wind and flood actions and achieves closed-loop interaction from “physical sensing” to “digital response”, providing a novel technical pathway for early damage identification and health monitoring of transmission towers in multi-hazard environments.

Compared to traditional FEM–IoT approaches that primarily output isolated numerical data, the proposed framework seamlessly maps real-time mechanical responses onto 3D BIM. This integration transforms abstract structural analyses into a spatially aware digital twin, enabling intuitive and multi-dimensional safety assessments under complex hazard scenarios [20,21].

2. Methodology

To enhance the accuracy and rapid response efficiency of the proposed digital twin framework, mechanical models accurately simulating wind loads and flood scour effects are established through FEM programs, employing wind load models, flood scour models, and soil–structure interaction (SSI) models recommended by international standards.

2.1. Wind Load Model

The wind load standard established in this paper follows international general standards [22,23], with the specific formula shown in Equation (1), which can be used as the basis for static wind load calculations for the high-rise transmission tower structure.

$$W_x = W_0 \mu_z \mu_s \beta_z B A_s \quad (1)$$

where W_x is the characteristic value of wind load (kN/m^2); W_0 is the basic wind pressure (kN/m^2); μ_z is the wind pressure height variation coefficient; μ_s is the wind load shape coefficient; β_z is the wind-induced vibration coefficient at height z , which equivalently simplifies the original complex random vibration wind load theory into a static load; B is

the ice accretion increase coefficient; and A_s is the projected area of the component on the windward side.

To reflect the random characteristics of the wind field in the FEM framework, the Davenport function spectrum [24] is introduced. By inputting the Davenport function spectrum through the Open Application Programming Interface (OAPI), the average wind speed can be transformed into a fluctuating wind speed time-history curve conforming to the actual wind field. This converts static wind speed calculations into dynamic wind load time-history functions, enabling dynamic analysis in FEM, which better aligns with the actual working conditions of high-rise structures subjected to wind loads. The specific expressions are shown in Equations (2) and (3).

$$S_z(n).dn = 4.0K\bar{V}_1^2 \cdot \frac{X}{(1 + X^2)^{4/3}} \cdot dx \quad (2)$$

$$X = \frac{1200n}{\bar{V}_1} \quad (3)$$

where $S_z(n)$ is the power spectral density function of the along-wind fluctuating wind speed; $\bar{V}_1$ is the mean wind speed at a height of 10 m (in m/s); n is the frequency of wind fluctuation (in Hz); X is the dimensionless frequency, used to normalize the spectral shape; and K is the ground roughness coefficient.

2.2. Flood Scour Mechanism and Simplified Foundation Stiffness Model

The transmission tower investigated in this study is located in a mountainous region prone to river scour. The foundation soil characteristics can be assumed as ideal non-cohesive sand. Therefore, the CSU (Colorado State University) formula from HEC-18 [25] is adopted, as shown in Equation (4), which can be applied to simulate specific flood scour scenarios for mountainous transmission towers.

$$\frac{y_s}{y_1} = 2.0k_1k_2k_3 \left(\frac{a}{y_1} \right)^{0.65} (Fr_1)^{0.43} \quad (4)$$

where y_s = scour depth (m); y_1 = flow depth directly upstream of the pier (m); k_1 = correction factor for pier nose shape; k_2 = correction factor for angle of attack of flow; k_3 = correction factor for bed condition; a = pier width (m); and Fr_1 = Froude number directly upstream of the pier = $V_1/(gy_1)^{1/2}$.

In this study, “ a ” in Equation (4) represents the diameter of a single-leg foundation of the transmission tower. The values of k_1 to k_3 are selected according to the definitions in HEC-18. Specifically, k_1 is taken as 1.0 for circular tower foundations and 1.1 for square foundations. Regarding k_2 , for circular foundations where the issue of flow attack angle is not considered, $k_2 = 1.0$. For the coefficient k_3 , when simulating mountain flood scour, $k_3 = 1.1$ can be adopted for flatbed riverbed conditions. Consequently, Equation (4) can be modified to Equation (5), which serves as the functional input for the flood scour model within the FEM framework of the digital twin proposed in this paper.

$$\frac{y_s}{y_1} = 2.2 \left(\frac{a}{y_1} \right)^{0.65} (Fr_1)^{0.43} \quad (5)$$

For mountainous transmission tower structures, the soil–structure interaction (SSI) model can refer to the p - y curves for sand formula recommended by API specifications [26], as shown in Equation (6). Reference [27] demonstrates that the chosen Winkler spring model is reliable, idealizing the soil as a series of discrete p - y springs, allowing the simulation of scour by simply reducing or removing spring elements within the scour depth. Refer-

ences [28,29] prove that scour can reduce the natural frequency of foundations. According to Equation (7), it can be concluded that flood scour reduces the structural foundation stiffness. Therefore, conducting flood scour research on transmission tower foundations through spring stiffness reduction has been proven reasonable and reliable. Furthermore, the parameters of this flood scour model can be introduced into FEM programs for flood scour simulation through the OAPI interface via functions, greatly improving computational efficiency compared to solid element models.

$$p = AP_u \tanh\left(\frac{kz}{AP_u} y\right) \quad (6)$$

where p = soil resistance per unit length (spring force per unit length); y = lateral deflection (m); k = initial modulus of subgrade reaction, which increases linearly with depth z ; z = depth below the soil surface (m); P_u = correction factor for bed condition, (kN); and A = factor to account for cyclic or static loading, evaluated as $A = 0.9$ for cyclic loading.

$$k = m\omega^2 \quad (7)$$

where k = stiffness of the structure foundation; m = mass of the structure; and ω = natural frequency of the transmission tower foundation structure.

The above wind load model and flood scour model respectively describe the load characteristics and boundary evolution laws of transmission towers under two typical types of disasters. In the digital twin framework constructed in this paper, these two types of models drive the finite element analysis separately through the OAPI interface, achieving parallel simulation of wind-induced vibration and foundation scour, providing the mechanical basis for the subsequent structural response analysis.

3. Digital Twin System Framework

3.1. Overview of Digital Twin Architecture

To clearly present the workflow of the digital twin framework in this paper, its working flowchart is shown in Figure 1. The main structure of the digital twin framework consists of three parts: Physical Entity Domain (Part 1), FEM Simulation Domain (Part 2), and Data and Control Center (Part 3). These three parts work together through the FEM Open Application Programming Interface (OAPI) and Python code programs. Each part works independently while being interconnected. Specifically, the FEM program from CSI (USA) is used to establish the finite element model of the transmission tower and complete structural calculations. Subsequently, the validated model data are imported into Revit 2025 BIM software to automatically generate a three-dimensional solid model, thereby using the BIM model as the visual window for terminal monitoring. The OAPI program acts as a “connector” in this process, ensuring efficient collaborative work among different application programs. Additionally, FEM calculation results can be mapped to the BIM program through the OAPI to establish a visual warning mechanism, achieving real-time monitoring effects. The system can automatically identify and capture abnormal data based on set thresholds and highlight risky structural components through rendering, achieving visual warning effects. The Python program acts as a binding agent in this process, enabling API call transmission work. It can achieve automated workflows through code language, including opening the FEM, setting analysis cases, extracting calculation results, and outputting them to data processing programs such as Excel, reducing the cumbersome processes of traditional finite element operations, improving data operation and processing efficiency, and achieving real-time interaction between the physical tower and the finite element model.

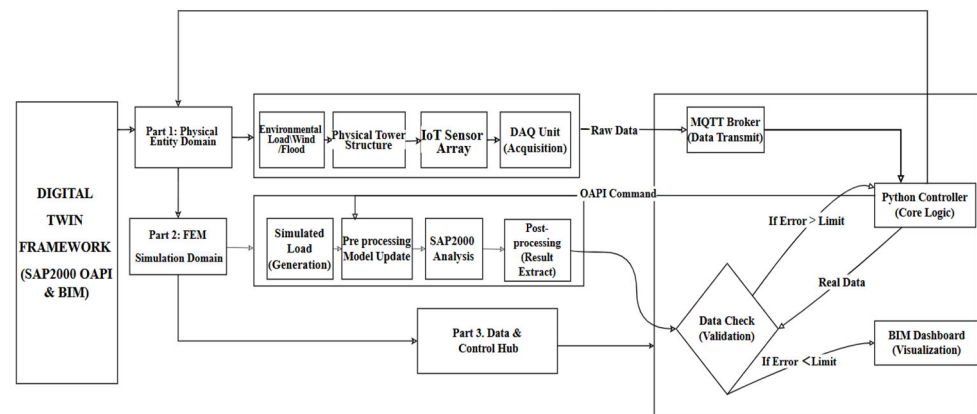


Figure 1. Flowchart of the proposed OAPI-based simulation methodology.

As shown in Figure 1, the digital twin framework proposed in this paper consists of three core domains: the Physical Entity Domain is responsible for environmental load generation and input; the Finite Element Simulation Domain undertakes the calculation and analysis of structural responses; the Data and Control Center Domain achieves cross-platform data transmission and command interaction. The three domains achieve closed-loop through the OAPI and Python scripts, forming a complete “sensing–calculation–feedback” digital twin chain.

To contextualize the proposed digital twin (DT) framework, Table 1 presents a comparison with existing structural analysis methods. While traditional Finite Element Methods (FEMs) and rigorous model updating serve as essential analytical foundations, the proposed framework builds upon these techniques by achieving a highly automated integration of physical modeling, continuous data orchestration, and BIM-based intuitive visualization for multi-hazard tracking.

Table 1. Comparison of the proposed digital twin framework with existing methods.

Feature/Methodology	Finite Element Modeling (FEM)	Modal Analysis	Model Updating	Proposed DT Framework
Data Interaction	Predominantly static	Driven by real-world data	Periodic/offline integration	Near real-time loop
Physical Interpretation	Design-based	Global dynamics	Calibrated parameters	Dynamically mapped
Automation Level	Manual setup	Semi-automated/signal processing	Often involves iterative tuning	Highly automated via OAPI
Visualization	Numerical stress contours	Frequency plots/mode shapes	Matrix/numerical outputs	Spatiotemporally synchronized 3D BIM
Primary Focus	Initial design & prediction	Dynamic characteristic extraction	Baseline structural calibration	Dynamic tracking & intuitive warning

3.2. Physical Entity Domain: Simulation Environment and Load Generation (Part 1)

3.2.1. Fluctuating Wind Load Input

This study generates wind loads through numerical simulation. The mathematical model employs the Davenport function wind spectrum model introduced in Section 2.1 to generate random wind fields. For the high-rise transmission tower structure discussed in this study, turbulence characteristics at different heights, especially special locations such as the tower base and tower top, can be considered. Rewriting the differential expressions

of Equations (2) and (3) yields the Davenport wind spectrum model function expressions input into the OAPI, as shown in Equations (8)–(10):

$$\frac{dX}{dn} = \frac{X}{n} \quad (8)$$

$$S_z(n) = 4.0K\bar{V}_1^2 \frac{X^2}{n(1 + X^2)^{4/3}} \quad (9)$$

$$P(t) = \frac{1}{2}C_{P0}P_a(U_t)^2 \quad (10)$$

where C_{P0} is a quasi-steady pressure coefficient, and $P(t)$ is the instantaneous wind pressure.

Expanding U_t into its mean and fluctuating components, we obtain Equation (11):

$$P(t) = \frac{1}{2}C_{P0}P_a(\bar{U} + u'_t)^2 \quad (11)$$

where u'_t is the fluctuating wind speed.

$$F(t) = \frac{1}{2}C_D P_a(\bar{U} + u'_t)^2 A \quad (12)$$

where C_D is the drag coefficient for the overall structure, A is the projected area of structure to the wind direction, and $P(t)$ denotes the time-history of wind pressure.

The instantaneous wind pressure $P(t)$, derived from Equation (11), is transformed into concentrated nodal forces $F(t)$ based on the spatial distribution characteristics and the effective wind-exposed areas of the structure. These forces are subsequently applied to the discrete nodes of the finite element model to perform dynamic analysis.

For practical engineering applications, current standards often neglect the higher-order nonlinear terms of fluctuating wind speeds to simplify aerodynamic load calculations [30]. While this approach improves design efficiency to some extent, it also reduces design accuracy to a degree. To accurately capture the nonlinear aerodynamic characteristics of structures under extreme conditions, retaining the fluctuating wind speed is more conducive to improving calculation accuracy for high-rise structures—especially transmission tower structures—under extreme wind load environments. This study adopts the quasi-steady theoretical formula proposed in reference [31], as shown in Equation (11).

Wind pressure in the free wind field acting on the structure follows the quasi-steady assumption, converting the free wind field's wind pressure into structural nodal loads through power spectral functions [32]. As shown in Equation (12), it defines the transformation of the instantaneous wind pressure $P(t)$.

Figure 2 presents the entire process of the automated wind load simulation and time-history analysis based on the FEM OAPI. This is an automated process driven by OAPI programs and Python scripts, realizing the mapping process from environmental wind field input to structural dynamic response for the transmission tower structure.

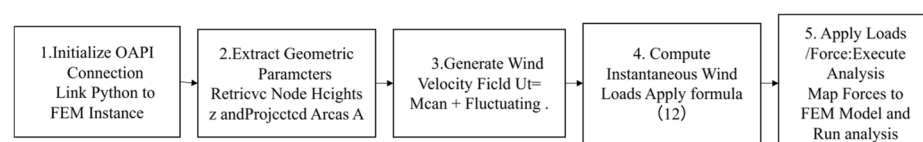


Figure 2. Automated wind load simulation and time-history analysis process based on FEM OAPI.

Step 1 extracts geometric parameters through the Frame Obj and Point Obj modules in the FEM OAPI program. In Step 2, the program groups by tower height, discretizing the structure along the tower height's direction, and applies nodal wind loads at different

heights. By implementing Equation (12) into the program, the structural stress conditions under extreme wind load conditions can be reflected. Finally, through Step 5, the Set Load Distributed module is driven for analysis and calculation.

3.2.2. Implementation of Flood Scour in OAPI and Python

To simulate the effects of mountainous flood scour on transmission tower foundations in the OAPI program, we adopt the simplified treatment of pile–soil interaction derived in Section 2 through Equations (5) and (6) in the Python program. Specifically, equivalent spring elements are used for the simulation at foundation nodes. Unlike the wind load application process, the flood scour model is mainly implemented by setting boundary conditions through the Set Multi Linear Points module in the OAPI program. Equation (5) can be used in the program to calculate scour depth, while Equation (6) calculates and updates the soil's support capacity for the tower foundation. By updating the spring link properties in the FEM OAPI program, an effect similar to a stiffness reduction caused by reduced soil around the foundation after scour is achieved, implemented in the program through the flowchart in Figure 3.

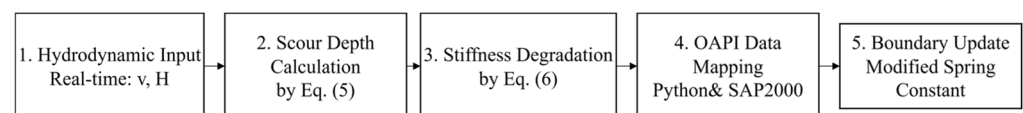


Figure 3. Flowchart of real-time scour-induced stress state assessment for transmission tower foundation based on OAPI.

3.3. IoT-OAPI-Based Finite Element Simulation Domain (Part 2)

In this study, the transmission tower is modeled using finite element (FE) software (SAP2000 V24). Virtual sensors are implemented by targeting specific node IDs to extract structural responses, such as nodal forces and accelerations, via a Python-based Open Application Programming Interface (OAPI). This approach effectively simulates the data output of physical sensors. Consequently, it establishes a direct linkage between the dynamic monitoring data and simulated disaster scenarios (e.g., flood-induced scour and strong winds), enabling an interactive evaluation of the tower's mechanical behavior under dynamic monitoring conditions.

In the visual warning interface, as shown in Figure 4, when simulating flood scour conditions, once the scour depth exceeds the set critical threshold, the boundary stiffness immediately degrades, thereby triggering internal force redistribution in the transmission tower foundation to simulate scour effects. The Python program automatically monitors the above mechanical responses through the OAPI interface. When monitoring those components, if stress ratios exceed safety limits, the system immediately identifies them as risky components, adds a red rendering display, and sends trigger instructions to the BIM. The corresponding digital elements in the BIM environment then automatically update to rendered bright-colored materials, providing intuitive structural health warnings. When simulating extreme wind load conditions, the transmission tower structure in FEM needs to be monitored according to control stress ratios and deflection limits. Through load data transmitted from the physical entity domain output and calculated by the FEM program, overall structural stress nephograms can be generated. When stress ratios are less than set limits or nodal displacements exceed code-specified limits, the same mapping mechanism as in flood conditions is triggered—corresponding components update to rendered material colors, providing warnings at the BIM visualization terminal. When wind loads and flood scour act simultaneously, the system makes judgments according to the envelope design, issuing warnings as soon as design requirements are not met.

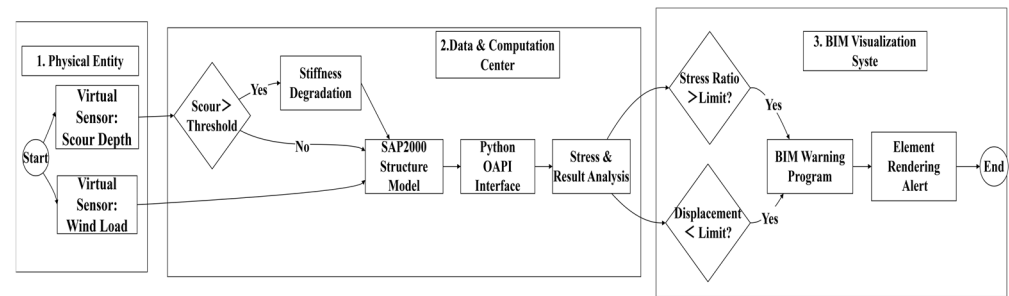


Figure 4. Early-warning mechanism at the BIM visualization terminal.

3.4. Data Control Center Domain (Part 3)

Serving as the core “connector” and “transporter” within the digital twin architecture, the Data and Control Center (Part 3) seamlessly bridges virtual sensor data streams with finite element (FE) simulation results. By leveraging a Python-based framework alongside the lightweight MQTT protocol, the center establishes a stable, asynchronous transmission topology tailored for real-time data delivery. Furthermore, the integration of advanced control logic and verification algorithms enables real-time compliance screening of incoming data via the FEM–Python–BIM visualization interface. Ultimately, this automated closed-loop system efficiently manages complex data routing while significantly reducing manual labor costs.

4. Finite Element Analysis

Addressing the pain point of feedback lag in traditional structural monitoring for disaster prevention and control, this paper proposes a digital twin interactive monitoring framework for transmission towers, aiming to achieve real-time synchronization of structural mechanical behavior and dynamic response. To validate this method, this study selected a widely used angle steel tower in power grids as a typical analysis object. Such high-rise structures are extremely sensitive to wind-induced vibrations, and the mechanical boundaries of their discrete members are very clear, highly compatible with the design method proposed in this paper. Through this model, research work was conducted, deeply integrating IoT sensing technology with three-dimensional information models and the finite element analysis software FEM. With the help of customized application programming interfaces, the cross-domain barrier from “physical-end IoT real-time data acquisition” to “virtual-end structural force deduction” was broken through, providing a new approach for early damage monitoring of transmission tower systems.

This study selected a lattice transmission tower as the research case, as shown in Figure 5. Specifically, the structure was divided into three main parts: the tower head, the tower body, and the foundation. In the head and body, all structural elements were made of L-shaped equal-leg angle steel. This included the main, diagonal, and auxiliary members. Furthermore, these steel members featured 20 different cross-sectional sizes, ranging from $L40 \times 3$ to $L140 \times 10$.

This study focused on a 44.4 m high latticed angle steel transmission tower. Specifically, the basic modeling parameters of this structure are detailed in Table 2. The primary material used was standard structural steel. In addition, the input material density was amplified to an equivalent mass density. This adjustment accounted for the weight of auxiliary components, such as gusset plates, bolted connections, climbing ladders, and anti-corrosion galvanized layers.

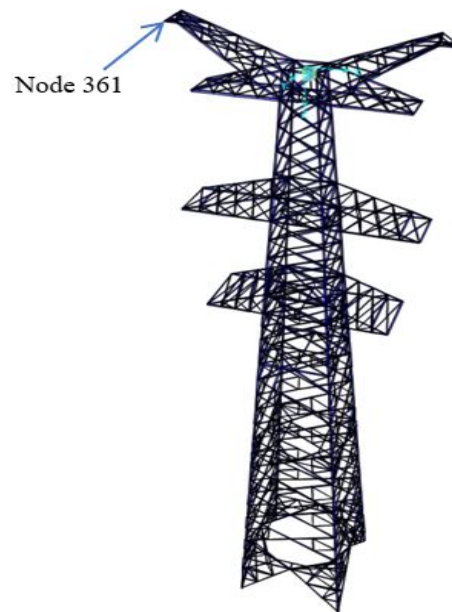


Figure 5. Transmission tower model.

Table 2. Structural Parameters of the Transmission Tower Model.

Parameters	Symbol	Value
Total height	H	44.4 m
Structural form	/	Lattice transmission tower
Elastic modulus	E	206 GPa
Poisson's ratio	γ	0.3
Base mass density	ρ_0	7850 kg/m ³

4.1. Simulation of Random Wind-Induced Vibration Effects on High-Rise Transmission Towers Under Complex Meteorological Environments

4.1.1. Fluctuating Wind Speed Simulation and Power Spectral Density Verification

This paper used the Davenport function from Section 3.2.1 as the theoretical basis for the frequency domain characteristics of wind loads. To meet the input requirements for time-history analysis, this paper simulated the time history of fluctuating wind speed at target nodes based on the harmonic superposition method. Input dynamic wind load parameters are shown in Table 3. The extracted wind speed time-history data underwent a fast Fourier transform using the Welch method [33,34] to obtain the auto-power spectrum (simulation) function curve, which was compared and verified with the Davenport theoretical target spectrum (theoretical), as shown in Figure 6. Results indicated that the simulated power spectrum curve closely matched the target power spectrum curve, proving that the power spectrum time-history function generated in this paper satisfied the random characteristics of natural wind in terms of energy distribution, providing a reliable load input boundary for subsequent OAPI structural time-history calculations.

As shown in Figure 7, the simulated wind speed fluctuates dramatically around the average wind speed of 36.1 m/s (red dashed line in the figure) at the height of the target node (Node 361), reflecting the randomness and fluctuating characteristics of natural wind. Figure 8 shows the dynamic wind load time-history curve generated by this fluctuating wind speed acting on the structural target node 361. The simulated spectrum closely matches the target Davenport spectrum in the main energy frequency band (0.1 Hz to 10 Hz), reflecting the high correlation between the fluctuation trend of dynamic wind force and the time-history characteristics of wind speed. To rigorously quantify the agreement

between the simulated and theoretical spectra, the Coefficient of Determination (R^2) was employed as a standard statistical metric [35,36]. The calculated R^2 value of 0.96 indicated a minimal error, demonstrating that the simulated stochastic wind field was highly accurate and perfectly suitable for the subsequent structural dynamic analysis.

Table 3. Basic parameters of the transmission tower model.

Parameters	Symbol	Value	Unit
Basic wind speed (10 m height)	V_{10}	26	m/s
Target simulation height	z	44.4	m
Ground roughness exponent	α	0.22	
Surface drag coefficient	K	0.005	
Simulation duration	T	600	s
Air density	ρ	1.225	kg/m ³

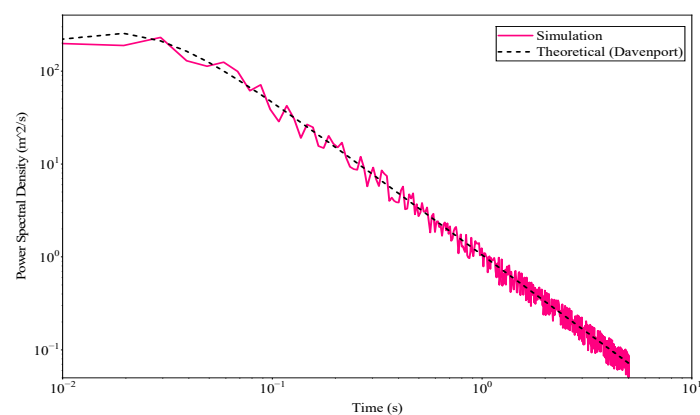


Figure 6. Comparison between simulated PSD and Davenport Spectrum.

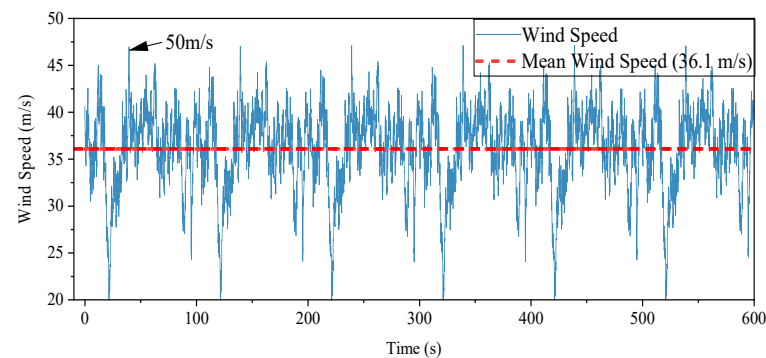


Figure 7. Time history of simulated wind speed and the corresponding dynamic wind force acting on target node 361 (duration: 600 s).

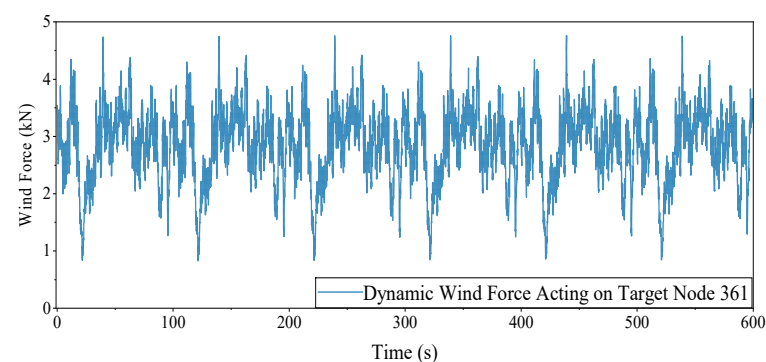


Figure 8. Dynamic wind force acting on target node 361.

From Figure 8, it can be seen that the 600 s generated time-history curve reflects the dynamic response of wind-induced vibration under actual environmental conditions, with actual wind speeds reaching 50 m/s—38.5% higher than traditional wind speeds. Therefore, it can be concluded that traditional static algorithms significantly underestimate the destructive power of transient gusts, and the dynamic wind load time-history analysis conducted in this paper can accurately capture this dynamic amplification effect. On the other hand, the traditional manual extraction mode is not only time-consuming and laborious but also prone to human operational errors. This study, based on the automated co-simulation of Python and FEM OAPI programs, achieves seamless data flow from environmental excitation to physical model to response output. After receiving the Davenport fluctuating wind sequence, the system can autonomously drive the solver to perform a nonlinear time-history integration and capture continuous high-frequency time-history responses (such as 6000-step displacement waveforms) of key nodes at high throughput. All processes are automatically completed through programs, reflecting the core essence of digital twin technology: real-time evolution and efficient data-driven operation.

4.1.2. Structural Dynamic Response Time-History Analysis

To verify the data interaction process of the digital twin framework, this paper used the wind force time-history curve generated in Section 4.1.1 as environmental excitation. These load data were published in real time as a continuous data stream relying on the MQTT protocol. Simultaneously, the FEM automatically read the above time-series data through the OAPI interface, thereby dynamically driving the structure to conduct a wind-induced vibration time-history analysis and subsequent flood scour analysis. Furthermore, for accurate comparison, “virtual sensors” were set at typical nodes at the tower base. These nodes strictly corresponded spatially to the IoT monitoring sensors of the on-site physical structure, and their main function was to synchronously extract and calculate the dynamic displacement responses and stress data of the tower in real time within the numerical model.

The wind-induced time-history curves generated in Section 4.1.1 were applied to the FEM model to conduct the dynamic time-history response analysis. To ensure the real-time computational efficiency of the proposed digital twin framework within its closed-loop interaction, geometric nonlinearities (P- Δ and large displacement effects) were not considered in this analysis. Existing studies [37] have also demonstrated that the influence of geometric nonlinearity on the dynamic characteristics of transmission towers is negligible in such cases. Consequently, obtaining the dynamic response solution without superimposing geometric nonlinearity effectively satisfied the low-latency requirements of the digital twin system while maintaining the requisite accuracy for stress and displacement calculations.

The wind force time-history curve from Section 4.1.1 was applied to the FEM model. This step facilitated a dynamic time-history analysis of the wind load. Specifically, it investigated the tower’s displacement patterns under wind-induced vibration. For this analysis, three representative nodes were selected: the tower base (Node 80), the body (Node 400), and the top (Node 361). Subsequently, their displacement responses were extracted and plotted as time-history curves, as illustrated in Figure 9.

In the figure, the horizontal displacements of each node increase with the height of the input tower section, growing from 8.9 mm to 90 mm. The tower top structure swings significantly, with the tower top node displacement reaching 90.24 mm, still meeting the deflection limit requirement of 3/1000 for self-supporting towers specified in the code [38]. The curves are highly correlated with the wind speed time history, exhibiting fluctuation characteristics consistent with the wind spectrum function features. Node 80, located near

the tower base, shows very small lateral displacement due to the strong constraint of its foundation, with the structure's bottom tending to be stable.

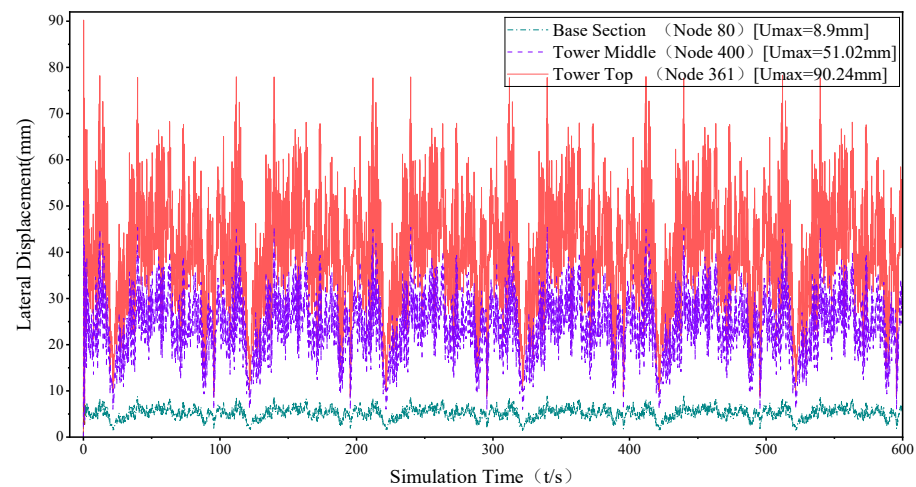


Figure 9. Multiscale dynamic response under design wind load.

Figures 10 and 11 show the first three natural frequencies and mode shapes of the transmission tower structure. The first mode is primarily an X-direction translation, the second mode is primarily a Y-direction translation, and the third mode is torsion about the Z-axis. The frequency deviation between the first and second modes is extremely small, indicating that the structure's stiffness distribution in the two translational directions (X and Y) is very uniform. The third-mode frequency increases by 58.5% compared to the first two, showing a stepwise jump in frequency distribution, indicating that the structure's torsional stiffness is greater than its translational stiffness.

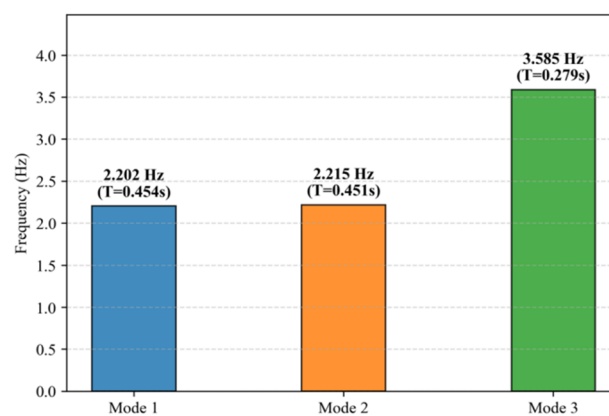


Figure 10. The first three natural frequencies.

Figure 12 shows the power spectral density (PSD) curve of the along-wind displacement response of the transmission tower model, with the first three natural frequencies of the structure added as reference lines. It can be clearly seen from the figure that the PSD curve has a very significant narrow-band energy peak around 2.2 Hz. Comparing this peak with the modal analysis results reveals that it accurately corresponds to the first (2.202 Hz) and second (2.215 Hz) natural frequencies of the structure. This frequency domain characteristic strongly proves that the fluctuating wind load excites the resonance of the structure's low-order modes, resulting in significant wind-induced resonant amplification effects. Meanwhile, this phenomenon does not occur at the third frequency, indicating that higher modes contribute little to the displacement response of this model.

Therefore, the dynamic response of this structure is mainly controlled by the first and second low-order modes.

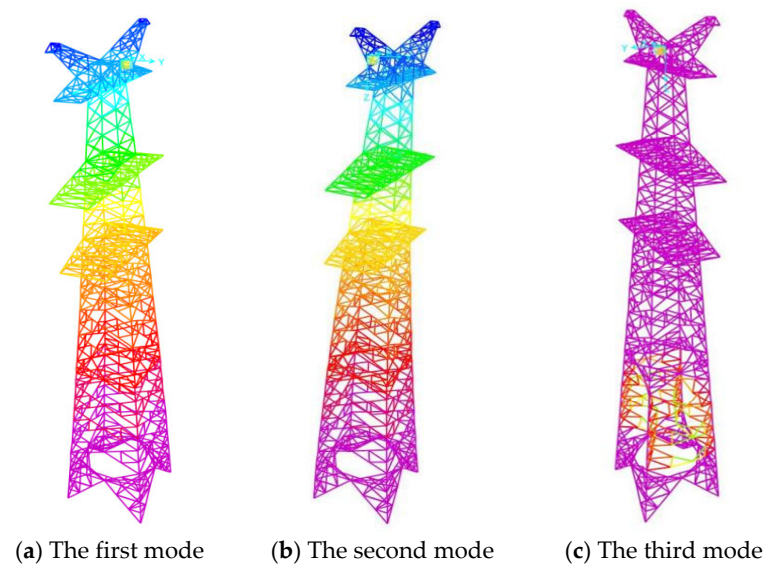


Figure 11. The first three formations.

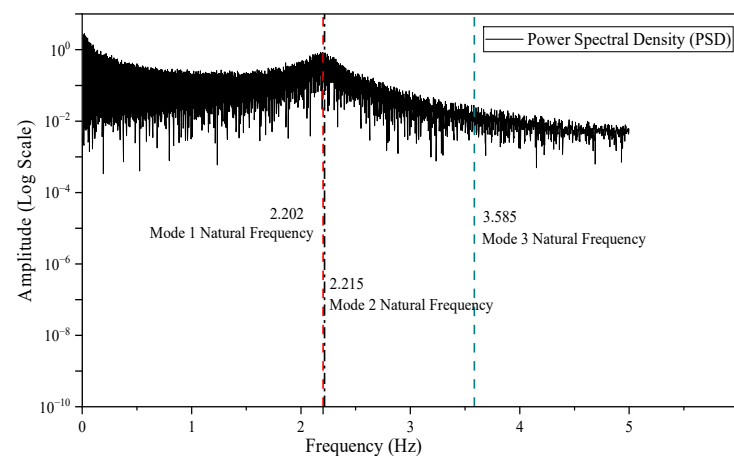


Figure 12. Frequency-domain correlation analysis.

4.2. Foundation Stiffness Attenuation and Nonlinear Evolution of Boundary Conditions Induced by Flood Scour

When conducting numerical modeling in the Python environment, the determination of foundation stiffness must consider the effects of scour on the structure. The classical API specification p-y curve model (as shown in Equation (6)) reveals that the lateral soil resistance of the foundation is positively correlated with soil depth z . When flood scour occurs, the surface soil is directly stripped away under flood loads, which not only causes a loss of rigid constraint at the tower foundation but also leads to a sharp decrease in the effective confining pressure of the underlying soil at the foundation burial depth. Due to the reduction in calculated depth z , the soil undergoes accelerated softening, exhibiting significant nonlinear attenuation characteristics.

Reference [39] indicates that when scour exceeds a critical depth, it triggers abrupt changes in soil–structure interaction boundaries, leading to nonlinear accelerated decline in structural natural frequencies. Reference [40] shows that severe scour can lead to almost complete loss of lateral foundation constraints, subsequently causing rigid body rotation. The above literature collectively demonstrates that traditional linear stiffness reduction

methods cannot truly simulate the cliff-like abrupt changes in boundary conditions during the later stages of scour.

Given the high sensitivity of high-rise transmission towers to overturning stiffness, this study nonlinearly mapped the degree of scour to an exponential reduction in equivalent spring stiffness, as shown in Equation (13), thereby realistically reproducing the evolution of the foundation from “elastic fixity” to a “near-hinged state” in the finite element environment. Through this approach, this study simulated the entire process of the transmission tower from local instability to eventual collapse, aiming to reveal the nonlinear dynamic evolution laws of high-rise structures under extreme scour conditions. The resulting flood scour parameter table is shown in Table 4.

$$K = K_0 \cdot 10^{-\alpha S_c} \quad (13)$$

where K = Equivalent Foundation Stiffness, representing the translational and rotational spring values under scouring in FEM. K_0 = initial reference stiffness, representing the foundation stiffness under unscoured conditions. Initial value: $K_0 = 10^7$ kN/m. α = decay rate factor. S_c = equivalent scour level, a dimensionless scalar ranging from zero to one (denoted as “level” in the code). $S_c = 0$: no scour occurred. $S_c = 1$: ultimate scour state.

Table 4. Stiffness parameter settings for foundation scour simulation.

S_c	Value	K_0 (kN/m)	Physical State Description
0%	$10^7 \times 10^0$	10^7	Initial healthy state, foundation fully fixed
20%	$10^7 \times 10^{-1}$	10^6	Soil loosening, stiffness degraded by one order of magnitude
40%	$10^7 \times 10^{-2}$	10^5	Severe scour, constraint capacity significantly reduced
60%	$10^7 \times 10^{-3}$	10^4	Critical state, frequency begins to exhibit cliff-like drop
80%	$10^7 \times 10^{-4}$	10^3	Failure mutation, foundation approaches “hinged” state, structure on the verge of collapse

The nonlinear mapping in Equation (13) is grounded in the stress-dependency theory of soil stiffness. According to [41], the removal of overburden soil layers during scour leads to a catastrophic reduction in effective confining pressure, causing a nonlinear degradation of the soil’s reaction modulus.

In the FEM model, the initial reference stiffness K_0 was set to 10^7 kN/m. This value was determined based on the standard parameters for elastic fixity provided in [42]. It ensured sufficient overturning resistance in the un-scoured state. The dimensionless scalar S_c (ranging from zero to one) defines the evolution from the pristine to the ultimate scour state. The decay coefficient α captures the geotechnical sensitivity of the site, simulating the transition from a rigid constraint to a near-hinged condition. To align with the API p-y framework, foundation degradation was modeled by attenuating both translational (U_1 , U_2 , U_3) and rotational (R_1 , R_2 , R_3) stiffness. Using nodal springs in FEM instead of solid elements ensured soil–structure interaction (SSI) fidelity while maintaining high computational efficiency.

In natural rivers or floods, scour never occurs uniformly; the scour experienced by foundations often exhibits certain randomness and non-uniformity. To determine the influence patterns of different scour distribution modes on transmission tower stability in the simulation, we designed two comparative working conditions—uniform scour and non-uniform scour—based on the above exponential degradation model. Under uniform scour

conditions, the stiffness of the four support nodes of the transmission tower foundation underwent exponential reduction synchronously with scour depth, aiming to simulate the idealized extreme state of overall riverbed down-cutting. Under non-uniform scour conditions, the upstream tower foundation nodes (45 and 47) were set to maintain rapid degradation with $\alpha = 5$, while the downstream tower foundation nodes (49 and 51) adopted a slower decay rate factor $\alpha = 3$, simulating situations with weaker local circulation.

From Figure 13a, it can be seen that when the scour degree is less than 40%, the first and second natural frequencies decrease relatively gently. When the scour degree exceeds 40%, the first and second natural frequencies of the tower show a nonlinear rapid decline trend, with a maximum reduction rate reaching 80.9%, greatly weakening the foundation fixity capacity. Under non-uniform scour, as shown in Figure 13b, although the frequencies at the upstream tower foundation still decrease rapidly, the natural frequency degradation at the downstream tower foundation is relatively slower, and the maximum reduction rate improves, ultimately decreasing to 73.8%.

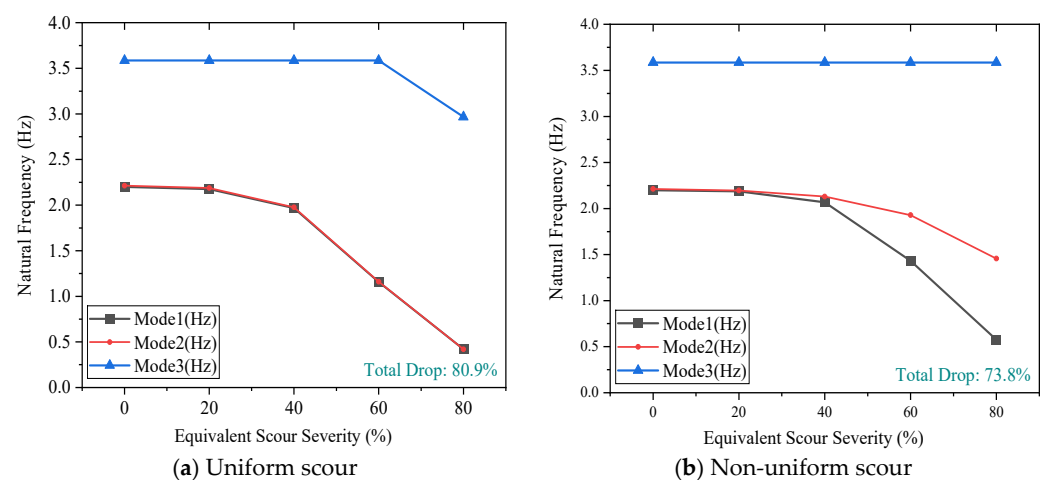


Figure 13. Comparison of modal frequency evolution of transmission tower under uniform and non-uniform scour conditions.

Overall, the first and second-order frequencies are greatly reduced under both uniform and non-uniform scour conditions, which is the result of the order-of-magnitude reduction in foundation spring stiffness. The first and second modes represent the bending moment effects of the structure in the horizontal direction, and the form of fixity between foundations greatly influences them. When stiffness attenuates, the first and second modes are greatly affected. The third mode represents the torsional effect of the structure, reflecting the spatial stiffness distribution characteristics. Uniform scour and non-uniform scour almost simultaneously reduce the foundation stiffness, but the spatial stiffness distribution changes little, mostly in a state of simultaneous reduction. Therefore, the natural frequencies of the third mode under both working conditions in Figure 13 show a relatively stable distribution trend.

Figure 14 shows the redistribution of axial forces and bending moments in the structure induced by non-uniform scour. Observation reveals that the changes in axial force in Figure 14a and bending moment in Figure 14b occur synchronously. In the initial stage of scour, the axial forces at the four nodes 45, 47, 49, and 51 are basically the same at 27 kN, and the structure is in equilibrium. However, as scour intensifies, the stiffness of upstream nodes 45 and 47 decreases significantly, causing them to be unable to support the upper load. Since gravity must find new support points to achieve equilibrium, the load begins to transfer to downstream nodes 49 and 51. When scour reaches 80%, the axial forces at upstream nodes 45 and 47 almost disappear due to the sharp drop in foundation stiffness

caused by scour, while the axial forces at downstream nodes 49 and 51 increase from approximately 27 kN to 54 kN. This doubling phenomenon indicates that the downstream foundation bears the entire weight of the tower, imposing significant compressive stress on that side of the foundation.

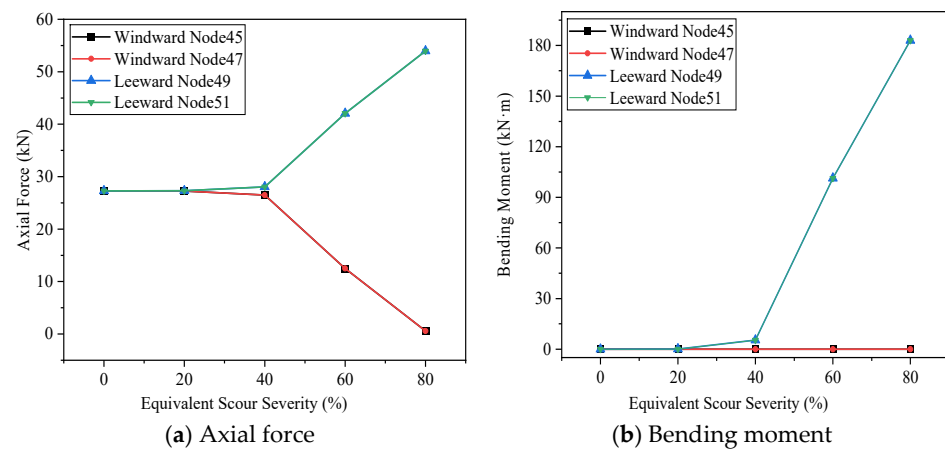


Figure 14. Internal force extraction of transmission tower under non-uniform scour conditions.

Furthermore, changes in base bending moments also reflect that the structure is in a dangerous state under scour. For upstream nodes 45 and 47, as the soil at the upstream foundation is gradually scoured, they cannot generate reaction forces to resist rotation, so the bending moment values are rapidly released and tend toward zero. This indicates that the upstream supports have lost their constraint capacity on the tower legs, changing from fixed to hinged. For downstream nodes 49 and 51, under non-uniform scour conditions, they need to anchor the tower body attempting to tilt. This strong constraint causes the bending moment values at nodes 49 and 51 to soar to 183 kN·m. This bending moment indicates that downstream nodes must not only resist compression but also bending, which can easily lead to “stiffness eccentricity”, causing the shearing of tower footing connection bolts or buckling of main members.

Further observation of Figure 13 reveals that when the equivalent scour severity (i.e., stiffness degradation) exceeds 40%, the natural frequency curves of the structure exhibit a distinct inflection point, followed by a sharp nonlinear decline. This evolutionary feature is highly consistent with established geotechnical dynamics. Reference [29] demonstrates that the natural frequency of a pile affected by scour does not decay linearly with scour depth; instead, it undergoes a sharp, nonlinear drop only after the scour reaches a critical threshold. Based on that reference and the curve mutation observed in Figure 13, this study identified 40% as the critical evolutionary threshold for the dynamic degradation of the transmission tower foundation under scour.

Figure 15 shows that non-uniform scour induces significant internal force transformation effects. On the one hand, at the axial force level, upstream and downstream nodes exhibit highly symmetrical force transfer. As scour increases from 20% to 80%, the load contribution rates show a “scissors difference” evolution (from 25% shifting to 24.3% and 25.7% respectively), reflecting the dangerous migration of several tons of gravity load to the safe side. On the other hand, at the bending moment level, redistribution is even more pronounced: the contribution rate of upstream foundations plummets below 10% in the early stage of scour (20%) due to the loss of rotational constraint, almost losing their role as fixed supports. When scour reaches 80% depth, the two downstream supports together bear nearly 100% of the bending moment load of the entire tower. Under this condition, the downstream foundation is more prone to failure and overall collapse. Due to the loss of stiffness at upstream supports (nodes 45, 47) causing an internal force release, the load is

forced to concentrate on downstream supports (nodes 49, 51), inducing additional bending moments up to 183 kN·m. This significant “axial force–bending moment” coupling effect can easily lead to the shearing of anchor bolts or instability of main members at downstream transmission tower foundations. Therefore, disaster prevention monitoring should focus on identifying the risk of force imbalance caused by local non-uniform scour, rather than only focusing on overall foundation settlement.

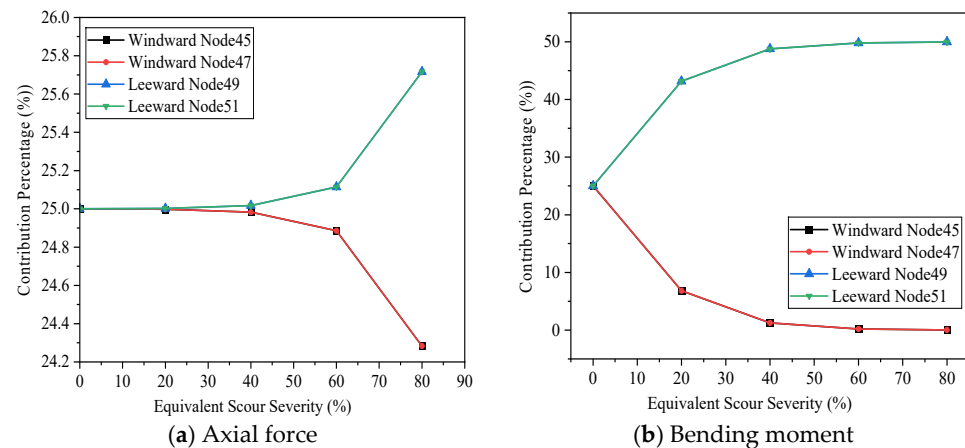


Figure 15. Evolution of internal force contribution rates under non-uniform scour conditions.

In summary, transmission towers are highly sensitive to foundation scour patterns. Under uniform scour conditions considering only gravity loads, the internal forces at each node remain stable with symmetric stiffness degradation, with risks mainly manifested as reduced overall bearing capacity while internal forces remain constant. Sections 4.1 and 4.2 respectively reveal the dynamic response characteristics under wind loads and the stiffness degradation laws induced by flood scour. Based on the above single-hazard analyses, Section 4.3 further demonstrates how to integrate and present structural response data from the two types of hazards through the BIM platform, achieving visual warning of structural health status under multi-hazard scenarios.

4.3. FEM–BIM Result Mapping and Dynamic Visual Warning of Structural Risks Based on OAPI

This study constructed an automated analysis pipeline across FEM–BIM platforms. First, FEM raw data were preprocessed in the Python environment. The Pandas library in the built-in Spyder program was used to extract stress ratios of each component and convert them into lightweight JSON format data. Subsequently, the Revit API was called in the Revit Python Shell plugin environment. External JSON databases were read through the interface to achieve precise alignment of component IDs between the FEM model and the BIM physical model, as shown in Figure 16.

A three-level visual mapping system was constructed by introducing the stress change rate (ΔR). A red zone represented areas with large stress changes under foundation displacement; a dark green transition zone depicted areas with significant internal force redistribution, reflecting the dynamic conduction process of stress changes from the tower bottom upward; a light gray stable zone used high transparency to hide information of components less affected by stress changes. The specific judgment criteria were carried out according to Table 5.

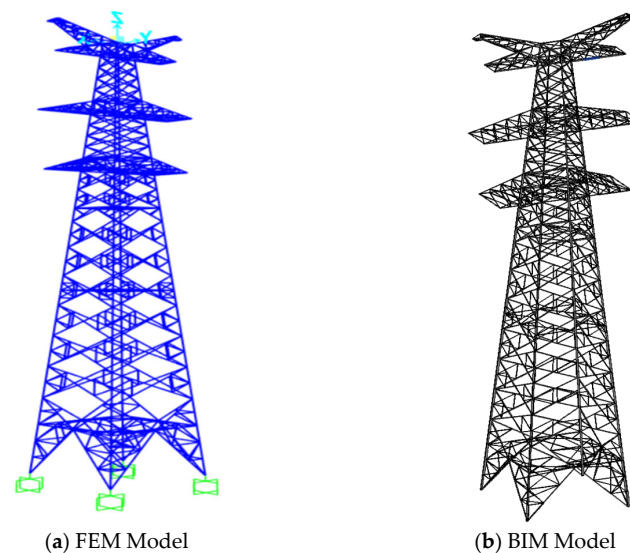


Figure 16. Bidirectional information conversion interaction between structural calculation model (FEM) and Building Information Model (BIM).

Table 5. Key mechanical parameter settings for foundation scour simulation.

Sensitivity Level	Criterion (ΔR)	Visual Mapping Scheme	Mechanical State Description
Highly Sensitive Mutation Zone	$\Delta R > 50\%$	Red	Stress outbreak point; local stiffness severely disturbed
Sub-Stable Transition Zone	$20\% < \Delta R < 50\%$	Dark green	Stress redistribution path; internal forces continuously increasing
Static Stable Zone	$0\% < \Delta R < 20\%$	Light gray	Structural steady-state zone; minimally affected region

In the initial stage of scour, most components of the tower body are in a gray hidden state representing stability, as shown in Figure 17a. This indicates that the additional internal forces generated by slight foundation displacements are still concentrated at the bottom support nodes and have not yet been transmitted on a large scale to the upper structure. In the late stage of scour, the structure exhibits color characteristics of increased stress change rates at multiple points. As shown in Figure 17b, the red sensitive zone not only locks onto the bottom failure points but also forms a connecting upward trend in the middle section of the tower body. This indicates that the geometric nonlinear effects caused by uneven settlement have led to the reorganization of the stiffness distribution throughout the tower, with the structure gradually transitioning from local disturbance to transmission to the upper structure.

During the entire 0–60 s wind loading process, the structural dynamic response exhibits significant stage-by-stage evolution laws with the increase in wind-load time history. Initially, in the low-wind-speed stage, the overall structure is in the static stable zone, with indicator $\Delta R < 20\%$. At this time, wind load disturbance to components is minimal, with only the tower leg's main members bearing the overall bending moment caused by wind load on the entire tower. The overall mechanical performance remains stable, mostly appearing gray and dark green in the BIM mapping model, as shown in Figure 18a. Subsequently, as the load intensity continues to rise, stress begins to spread along specific paths, entering the sub-steady transition zone ($20\% < \Delta R < 50\%$). From

a mechanical logic perspective, this stage constitutes a stress redistribution path, with internal force conduction becoming increasingly active, playing a key role in load transfer and energy dissipation. At this time, the stress distribution on main members gradually spreads to the upper structure, as shown in Figure 18b. Finally, at the 60 s moment reaching the load peak, the red range in the tower leg area further expands, i.e., the stress burst point with $\Delta R > 50\%$ becomes the weakest link in the overall structural wind resistance, as shown in Figure 19a. In summary, the BIM mapping results accurately capture the entire process of structural stress evolution, and the FEM stress nephogram at the 60 s moment highly agrees with the damage indicator ΔR , as shown in Figure 19b, achieving closed-loop completion of the digital twin process.

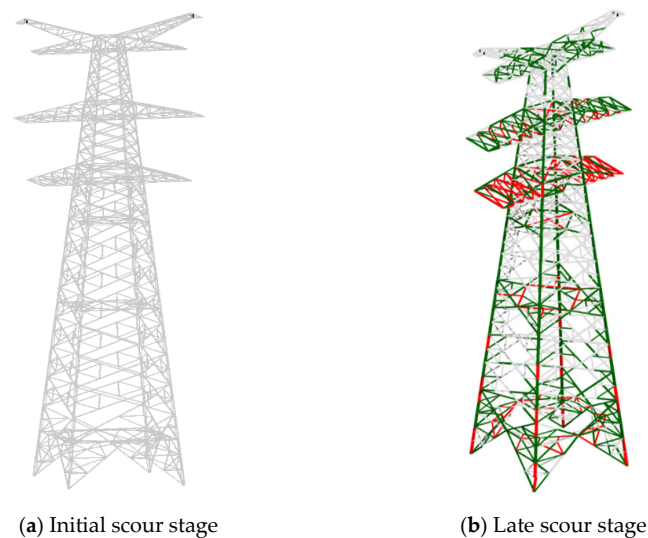


Figure 17. BIM model mapping response under foundation scour.

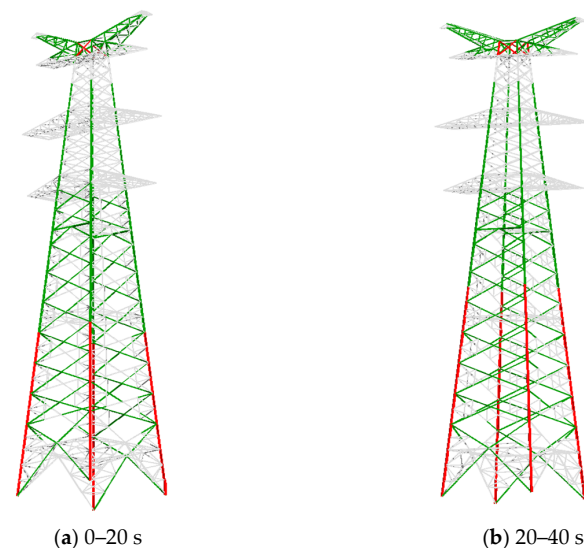


Figure 18. BIM model mapping response under flood scour.

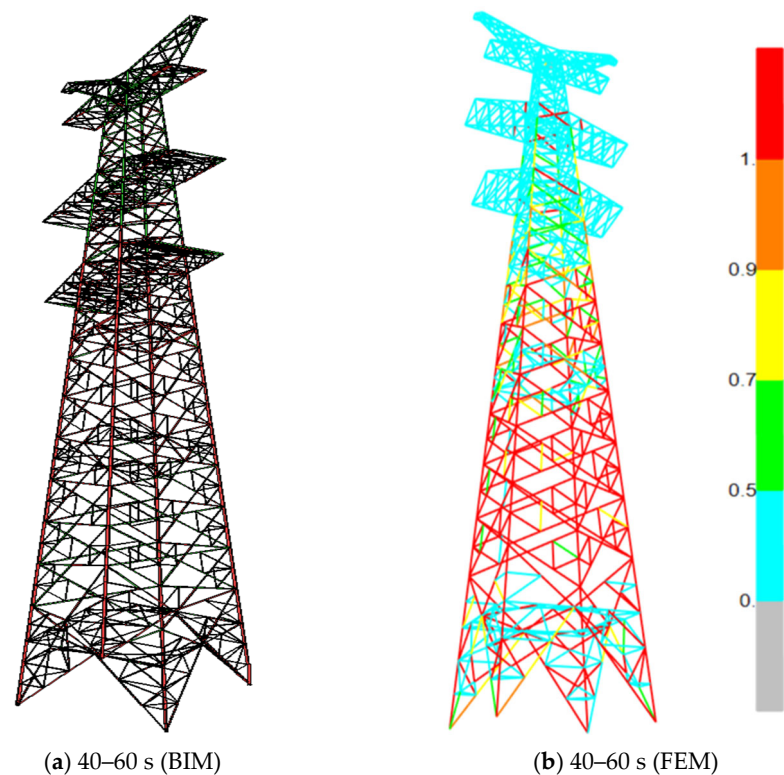


Figure 19. Comparative mapping response of BIM and FEM models under foundation scour.

5. Conclusions and Future Work

5.1. Conclusions

The digital twin framework proposed in this paper integrates IoT, BIM, and FEM. Through low-level calls of the OAPI interface in the Python environment, real-time data from the physical environment can directly drive the mechanical evolution analysis in the virtual space. This virtual–real interaction mechanism effectively enhances the timeliness of simulation deduction, providing innovative ideas for dynamic identification of early damage in transmission towers. Specific conclusions are as follows:

(1) The Davenport fluctuating wind spectrum was generated based on the harmonic superposition method, completely retaining the square term of wind load. Verified by the Welch method, the simulated power spectrum agreed well with the theoretical target spectrum. Results showed that this method accurately reproduced the random characteristics of natural wind. Meanwhile, it effectively overcame the limitations of traditional linear assumptions, providing reliable load boundary conditions for the OAPI-driven digital twin framework.

(2) A digital twin closed-loop mechanism driven by IoT data was successfully constructed. This study used the IoT monitoring network as the core information source, relying on MQTT and OAPI interfaces to efficiently convert real-time environmental data into dynamic inputs for the finite element analysis, driving structural wind-induced vibration and scour analysis. Simultaneously, by interconnecting model nodes with on-site IoT devices, a “virtual sensor” mapping was established. This framework ensured the synchronous analysis and calculation of stress and displacement responses, completing the data closed-loop mechanism from physical to digital ends.

(3) Through nonlinear exponential spring stiffness reduction, the entire process of dynamic degradation of transmission towers under flood scour was realistically reproduced. Results indicated that the structural natural vibration characteristics had a 40% scour evolution threshold. Structural frequencies exhibited significant stage-sensitive char-

acteristics to the degree of scour. After exceeding that threshold, the first and second frequencies showed a rapid nonlinear decay (maximum reduction of 80.9%), with the foundation rapidly degrading from a fixed end to a near-hinged state. Additionally, scour induced significant load transfer effects. When scour reached 80%, the bearing capacity of the upstream foundation basically failed; simultaneously, axial forces at downstream nodes surged, doubling from 27 kN to 54 kN.

(4) During the entire 0–60 s loading process, the structural dynamic response exhibited significant stage characteristics. In the initial stage of wind loading, the stress change rate was small ($0\% < \Delta R < 20\%$), the structure maintained stability, and the tower legs mainly bore the bending moment caused by wind load. In the middle stage of wind loading, the stress change rate further expanded ($20\% < \Delta R < 50\%$), with stress spreading upward and redistributing. At the peak moment of the wind-load time history reaching 60 s, the local stress change rate of the tower legs reached its peak, with $\Delta R > 50\%$, and the tower bottom became a stress concentration area, developing into a weak link in wind resistance. Finally, the FEM mechanical response highly agreed with the damage indicator ΔR , accurately mapping the evolution process of structural internal forces and successfully achieving closed-loop validation of the digital twin framework.

Practically, the strong structural asymmetry observed in this study suggests that future design and safety assessment frameworks should explicitly incorporate non-uniform scour as a critical load case. Accounting for this spatial variability helps to ensure that localized extreme internal forces and differential settlements are adequately addressed. Furthermore, structural health monitoring and maintenance strategies can be optimized by targeting these critical regions of structural degradation.

5.2. Future Work

(1) The current work primarily focused on validating the feasibility of the proposed digital twin framework. Consequently, the analyses were conducted for wind load and flood scour as separate, single-hazard scenarios. A comprehensive analysis of wind–flood joint actions will be the focus of subsequent research.

(2) The current framework demonstrates the integrated feasibility of automated multi-hazard tracking under idealized conditions, but it does not yet account for complex uncertainty propagation across the independent models (e.g., wind, scour, and SSI). Following the profound insights into probabilistic paradigms [43] and uncertainty quantification [44,45] established by recent pioneering works, future research will focus on quantifying and propagating these multi-source uncertainties through the digital twin framework. This will facilitate a crucial transition from deterministic visualization to robust probabilistic reliability assessment, thereby further perfecting the fidelity of the digital twin system.

(3) Future research will extend this framework to thermo-mechanical hazards, as emerging risks of electric fires with thermal runaway and structural degradation are increasingly critical trends in infrastructure resilience [46,47].

(4) While the current digital twin framework demonstrates strong numerical consistency through PSD matching and validation using virtual sensors, future research will focus on its real-world implementation. We plan to integrate physical IoT hardware (e.g., anemometers, sonar profilers, and strain gauges) to rigorously validate the dynamic twinning accuracy under complex, real-world scour scenarios using field-measured data and scaled laboratory experiments. Furthermore, to ensure practical scalability, future work will explicitly evaluate the cost-benefit of different sensor deployment schemes (e.g., sparse vs. dense networks) to optimize hardware and maintenance costs while maintaining high monitoring reliability.

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Conflicts of Interest: Authors Xiaoqing Qi, Huaichao Wang, Xiaoyu Xiong, and Angi Zhou were employed by the company Beijing Electric Power Economic and Technological Research Institute Co., Ltd, Beijing 100055, China. The remaining authors Qing Sun and Qiang Zhang declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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